# H524 Introduction to Biostatistics

## Week 2: Probability Concepts and Applications

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# Outline

# Quick Review: Key Concepts

- **Data Types:** Quantitative vs Qualitative
- **Data Presentation:** Histograms, box plots, bar charts
- **Summary Measures:** Central tendency and variability
- **Rates and Standardization:** Age-specific rates, direct/indirect methods

**This Week:** Chapter 6 - Probability foundations for statistical inference

**Reading:** Pagano & Gauvreau Chapter 6 (6.1-6.4.2, 6.5)

# Why Probability in Biostatistics?

- **Foundation for Statistical Inference**
  - Understanding uncertainty in medical research
  - Bridge between sample data and population conclusions
- **Clinical Applications**
  - Diagnostic test interpretation
  - Treatment efficacy assessment
  - Risk communication to patients
- **Research Design**
  - Power calculations
  - Sample size determination
  - Study planning and interpretation

# Basic Probability Concepts

- **Experiment:** Any process that generates observations
  - Example: Testing a patient for COVID-19
- **Sample Space ($S$):** Set of all possible outcomes
  - Example: $S = \{\text{Positive}, \text{Negative}\}$
- **Event:** Subset of the sample space
  - Example: Patient tests positive
- **Probability:** Measure of likelihood (0 to 1)
  - $P(\text{Event}) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$

# Probability Rules

1. **Range:** $0 \leq P(A) \leq 1$ for any event $A$
2. **Certainty:** $P(S) = 1$ (something must happen)
3. **Complement:** $P(A^c) = 1 - P(A)$
4. **Addition Rule:**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

5. **Multiplication Rule:**

$$P(A \cap B) = P(A) \times P(B|A)$$

# Medical Example: Diagnostic Testing

**COVID-19 Rapid Test Performance:**

- Sensitivity = 85% (probability of positive test given disease)
- Specificity = 97% (probability of negative test given no disease)
- Disease prevalence = 5%

**Questions:**

- What's the probability of a false positive?
- What's the probability someone with a positive test has COVID?
- How does prevalence affect test interpretation?

These questions require understanding conditional probability!

# Conditional Probability

**Definition:** Probability of event $A$ given that event $B$ has occurred

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \text{ where } P(B) > 0$$

**Medical Example:**

- $A$ = Patient has diabetes
- $B$ = Patient is overweight
- $P(A|B)$ = Probability of diabetes given patient is overweight

**Key Insight:** Conditional probability updates our beliefs based on new information

# Independence

**Definition:** Events $A$ and $B$ are independent if:

$$P(A|B) = P(A) \text{ or equivalently } P(A \cap B) = P(A) \times P(B)$$

**Examples in Healthcare:**

- **Independent:**
  - Blood type and eye color
  - Two different patients' test results
- **Dependent:**
  - Smoking status and lung cancer
  - Age and cardiovascular disease
  - Family history and genetic disorders

**Caution:** Independence is often assumed but should be verified!

# Diagnostic Test Performance

**Key Metrics for Medical Tests:**

- **Sensitivity:** P(Positive Test — Disease Present)
  - Ability to correctly identify disease
  - True positive rate
- **Specificity:** P(Negative Test — Disease Absent)
  - Ability to correctly identify no disease
  - True negative rate

**Example - COVID-19 Rapid Test:**

- Sensitivity $=$ 85% (detects 85% of infected people)
- Specificity $=$ 97% (correctly identifies 97% of non-infected)

# Predictive Values

**Clinical Interpretation Measures:**

- **Positive Predictive Value (PPV):** P(Disease — Positive Test)
  - Probability patient has disease given positive test
  - Depends on disease prevalence
- **Negative Predictive Value (NPV):** P(No Disease — Negative Test)
  - Probability patient is healthy given negative test
  - Also depends on disease prevalence

**Key Insight:** PPV and NPV change with disease prevalence in the population, while sensitivity and specificity are test characteristics.

# 2x2 Contingency Tables

| | **Disease +** | **Disease -** | **Total** |
|---|---|---|---|
| **Exposed** | a | b | a + b |
| **Not Exposed** | c | d | c + d |
| **Total** | a + c | b + d | n |

**Key Measures:**

- **Risk in Exposed:** $R_E = \frac{a}{a+b}$
- **Risk in Unexposed:** $R_U = \frac{c}{c+d}$
- **Relative Risk (RR):** $\frac{R_E}{R_U} = \frac{a/(a+b)}{c/(c+d)}$
- **Odds Ratio (OR):** $\frac{ad}{bc}$

# Relative Risk vs Odds Ratio

- **Relative Risk (RR):**
  - Direct measure of risk comparison
  - Used in cohort studies
  - RR $= 2$ means exposed have twice the risk
  - Cannot be calculated in case-control studies
- **Odds Ratio (OR):**
  - Approximates RR when disease is rare
  - Can be calculated in all study designs
  - More difficult to interpret directly
  - OR ¿ RR when outcome is common

**Rule of thumb:** When disease prevalence ¡ 10%, OR $\approx$ RR

# Example: Smoking and Lung Cancer

**Study of 1000 individuals over 20 years:**

| | **Lung Cancer** | **No Cancer** | **Total** |
|---|---|---|---|
| **Smokers** | 60 | 440 | 500 |
| **Non-smokers** | 10 | 490 | 500 |
| **Total** | 70 | 930 | 1000 |

**Calculations:**

- Risk in smokers: $60/500 = 0.12$ (12%)
- Risk in non-smokers: $10/500 = 0.02$ (2%)
- Relative Risk: $0.12/0.02 = 6.0$
- Odds Ratio: $(60 \times 490)/(10 \times 440) = 6.68$

**Interpretation:** Smokers have 6 times the risk of lung cancer

# Expected Value

**Definition:** Long-run average value of a random variable

**Discrete Random Variable:**

$$E(X) = \mu = \sum_i x_i \cdot P(X = x_i)$$

**Medical Example:** Number of adverse events in clinical trial

| Events ($x$) | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| Probability | 0.70 | 0.20 | 0.08 | 0.02 |

$$E(X) = 0(0.70) + 1(0.20) + 2(0.08) + 3(0.02) = 0.42$$

**Interpretation:** On average, expect 0.42 adverse events per patient

# Variance and Standard Deviation

**Variance:** Measure of spread around the expected value

$$\text{Var}(X) = \sigma^2 = E[(X - \mu)^2] = E(X^2) - [E(X)]^2$$

**Standard Deviation:** Square root of variance

$$SD(X) = \sigma = \sqrt{\text{Var}(X)}$$

**Properties:**

- $\text{Var}(aX) = a^2\text{Var}(X)$
- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ if $X$ and $Y$ are independent
- Standard deviation has same units as original data

# Binomial Distribution

**Conditions:**

- Fixed number of trials ($n$)
- Two possible outcomes (success/failure)
- Constant probability of success ($p$)
- Independent trials

**Probability Mass Function:**

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

**Parameters:**

- Mean: $\mu = np$
- Variance: $\sigma^2 = np(1 - p)$

# Binomial Example: Vaccine Efficacy

**Clinical Trial:** 20 vaccinated individuals exposed to virus

- Vaccine efficacy = 90% (probability of protection)
- What's the probability exactly 18 are protected?

**Solution:**

- $n = 20$, $p = 0.90$, $k = 18$
- $P(X = 18) = \binom{20}{18}(0.90)^{18}(0.10)^2$
- $P(X = 18) = 190 \times 0.1501 \times 0.01 = 0.285$

**Expected number protected:** $\mu = 20 \times 0.90 = 18$
**Standard deviation:** $\sigma = \sqrt{20 \times 0.90 \times 0.10} = 1.34$

# Normal Distribution

**Properties:**

- Bell-shaped, symmetric curve
- Characterized by mean ($\mu$) and standard deviation ($\sigma$)
- Area under curve $= 1$
- Approximately 68% within $\pm 1\sigma$
- Approximately 95% within $\pm 2\sigma$
- Approximately 99.7% within $\pm 3\sigma$

**Standard Normal:** $Z = \frac{X-\mu}{\sigma}$ has mean 0, SD 1

**Medical Applications:**

- Blood pressure measurements
- Cholesterol levels
- Height and weight
- Many lab values

# Normal Distribution Example

**Systolic Blood Pressure in Adults:**

- Mean = 120 mmHg
- Standard deviation = 15 mmHg
- What proportion have BP ¿ 140 mmHg (hypertension)?

**Solution:**

1. Standardize: $Z = \frac{140-120}{15} = 1.33$
2. Look up in Z-table: $P(Z < 1.33) = 0.9082$
3. Therefore: $P(BP > 140) = 1 - 0.9082 = 0.0918$

**Interpretation:** About 9.2% of adults have hypertension by this criterion

# Probability in Public Health

**Disease Surveillance Example:**

- Population: 50,000 individuals
- New disease cases: 125 per year
- Deaths from disease: 15 per year

**Calculate probabilities:**

- P(disease) = 125/50,000 = 0.0025 (0.25%)
- P(death — disease) = 15/125 = 0.12 (12%)
- P(death and disease) = 15/50,000 = 0.0003 (0.03%)

**Clinical interpretation:**

- Incidence rate: 250 per 100,000 per year
- Case fatality rate: 12%
- Overall mortality rate: 30 per 100,000 per year

# R Example: Basic Probability

```r
# Probability calculations in R
# Sample space for rolling two dice
dice1 <- 1:6
dice2 <- 1:6
sample_space <- expand.grid(dice1, dice2)
sample_space$sum <- sample_space$Var1 + sample_space$Var2

# Probability of sum = 7
p_seven <- sum(sample_space$sum == 7) / nrow(sample_space)
cat("P(sum = 7) =", p_seven, "\n")  # 6/36 = 1/6

# Conditional probability example
# P(sum > 8 | first die = 4)
subset_data <- sample_space[sample_space$Var1 == 4, ]
p_conditional <- sum(subset_data$sum > 8) / nrow(subset_data)
cat("P(sum > 8 | first = 4) =", p_conditional, "\n")  # 3/6 = 1/2
```

# R Example: Medical Testing

```
# Breast cancer screening example
sens <- 0.90  # Sensitivity
spec <- 0.95  # Specificity
prev <- 0.008 # Prevalence

# Calculate PPV and NPV
ppv <- (sens * prev) /
       (sens * prev + (1-spec) * (1-prev))
npv <- (spec * (1-prev)) /
       ((1-sens) * prev + spec * (1-prev))

cat("PPV:", round(ppv * 100, 1), "%\n")
cat("NPV:", round(npv * 100, 1), "%\n")

# Simulate population
n <- 10000
n_disease <- round(n * prev)
n_healthy <- n - n_disease

# Test outcomes
TP <- round(n_disease * sens)
```

# R Example: Binomial Distribution

```
# Clinical trial with 20 patients, treatment success rate 70%
n <- 20
p <- 0.7

# Probability of exactly 15 successes
p_15 <- dbinom(15, n, p)
cat("P(X = 15) =", round(p_15, 4), "\n")

# Probability of at least 15 successes
p_atleast_15 <- pbinom(14, n, p, lower.tail = FALSE)
cat("P(X >= 15) =", round(p_atleast_15, 4), "\n")

# Expected value and standard deviation
expected <- n * p
std_dev <- sqrt(n * p * (1-p))
cat("Expected successes:", expected, "\n")
cat("Standard deviation:", round(std_dev, 2), "\n")

# Visualize distribution
```

# R Example: Normal Distribution

```
# Blood pressure in adults
mu <- 120       # Mean SBP
sigma <- 15     # Standard deviation

# Proportion with hypertension (SBP > 140)
p_hyper <- pnorm(140, mu, sigma, lower.tail = FALSE)
cat("P(BP > 140):", round(p_hyper * 100, 1), "%\n")

# Find percentiles
bp_90 <- qnorm(0.90, mu, sigma)
bp_95 <- qnorm(0.95, mu, sigma)
cat("90th percentile:", round(bp_90, 1), "mmHg\n")
cat("95th percentile:", round(bp_95, 1), "mmHg\n")

# Visualize distribution
x <- seq(70, 170, length = 100)
plot(x, dnorm(x, mu, sigma), type = "l",
     main = "Systolic Blood Pressure",
     xlab = "SBP (mmHg)", ylab = "Density")
abline(v = c(mu, 140), col = c("blue", "red"))
```

# Practice Problems

1. **Diagnostic Test:** A rapid strep test has 85% sensitivity and 95% specificity. If 15% of patients with sore throat have strep, what's the NPV?
2. **Risk Calculation:** In a study of 1000 people, 200 smokers and 800 non-smokers, 30 smokers and 20 non-smokers developed heart disease. Calculate the relative risk.
3. **Binomial:** If a treatment works in 60% of patients, what's the probability that it works in at least 8 out of 10 patients?
4. **Normal:** If cholesterol levels are normally distributed with mean 200 mg/dL and SD 40 mg/dL, what proportion of people have levels between 180 and 220?

# Summary: Chapter 6 - Probability

**Today we covered:**

- **Section 6.1:** Basic probability concepts, rules, and definitions
- **Section 6.2:** Conditional probability and independence
- **Section 6.3:** Diagnostic test applications (sensitivity, specificity)
- **Section 6.4:** Random variables and expected value
- **Section 6.5:** Discrete distributions (binomial)

**Key Learning Objectives:**

- Calculate and interpret basic probabilities
- Understand conditional probability and independence
- Apply probability concepts to diagnostic testing
- Use probability distributions for modeling

**Next Week:** Chapter 7 & 8 - Theoretical distributions and sampling

# Questions?

Thank you!

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**Reading for next week:**
Pagano & Gauvreau, Chapter 7 (7.1, 7.2, 7.4) and Chapter 8